

Beyond Nested Sampling: A Survey of Samplers for Multimessenger Bayesian Inference

Research notes for the NMMA / retreat discussion

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Abstract

NMMA performs joint Bayesian inference over gravitational-wave (GW), electromagnetic (kilonova/GRB) and nuclear equation-of-state (EOS) data by wrapping `bilby` around nested samplers (`pymultinest` by default, plus `dynesty` and `ultranest`). This note surveys (i) which other codes in the multimessenger/nuclear-astrophysics space solve structurally similar problems, whether they are open source, and which samplers they use, and (ii) what neighbouring and more distant Bayesian-computation communities facing high-dimensional, expensive-likelihood inference have already adopted, with performance numbers checked directly against each paper’s own L^AT_EX source rather than quoted from abstracts. Headline finding: essentially the entire multimessenger/nuclear-EOS ecosystem (NMMA, `bajes`, `redback`, `NEoS`, the nonparametric-EOS lineage) still relies on the same CPU-bound nested-sampling engines, while the pure-GW subfield next door has already moved to GPU-native, gradient-based or amortized samplers with verified $5\times$ to $\sim 400\times$ wall-clock reductions on problems of comparable or higher dimensionality — including two pieces of work the group itself has already co-authored. This is a concrete, low-risk opportunity, not a hypothetical one.

1 The starting point: how NMMA samples today

A direct inspection of the NMMA source (<https://github.com/nuclear-multimessenger-astronomy/nmma>) confirms the following, rather than assuming it from the papers alone:

- NMMA’s inference stack is built on `bilby` [1], following the original and updated NMMA methodology papers [2, 3], with an extended framework paper released this year [4]. All of its supported samplers are CPU-bound nested samplers: `pymultinest` is the CLI default (MultiNest, [5]), `dynesty` [6] is a core dependency, and `ultranest` [7] ships as an optional extra (`nmma/core/parsing.py`, `pyproject.toml`).
- EOS inference solves the Tolman–Oppenheimer–Volkoff equations directly inside the likelihood: `nmma/eos/tov.py` integrates the stellar-structure ODEs with `scipy.integrate.solve_ivp` for *every* sampled EOS, at every mass on the mass grid, on every likelihood call — no caching, surrogate, or amortization across samples.
- Parallelism is handled purely at the level of MPI across independent likelihood evaluations (`nmma/core/mpi_setup.py`): more CPU workers, not GPU or gradient-based acceleration.

This is a reasonable default: nested sampling is robust to multimodality, needs no gradients, and directly returns the Bayesian evidence NMMA needs for model selection between EOS families or kilonova/GRB models. The cost is that nested sampling’s live-point evolution requires $\mathcal{O}(10^5\text{--}10^7)$ likelihood evaluations per run, growing with dimensionality and with the correlation length of the target. When the likelihood is itself slow — GW waveform generation, an on-the-fly TOV solve,

a kilonova/GRB light-curve evaluation — the sampler’s evaluation count directly sets the wall-clock budget, and joint multimessenger analyses stack several such slow likelihoods (and their parameters) at once. This compounds exactly where NMMA’s science ambitions (more sources, more parameters, next-generation-detector sensitivities) are pushing hardest.

2 Who else samples multimessenger sources, and how?

Searching INSPIRE-HEP for codes and papers performing Bayesian parameter estimation on GW, EM-transient, or EOS data in a way structurally comparable to NMMA turns up a small, well-defined set of tools, summarized in Table 1.

Code	Reference	Open?	Sampler(s)	Scope
NMMA	[2, 3, 4]	Y	pymultinest, dynesty, ultranest (via bilby)	Joint GW+EM+EOS
bajes	[8, 9]	Y	dynesty (previously cpnest)	Joint GW+kilonova
NEoS	[10]	Y	Nested sampling (MultiNest/dynesty-family)	EOS from NICER+GW
Nonparametric EOS (GP/spectral)	[11, 12, 13, 14]	Mixed	Nested/importance/rejection sampling over GP-drawn EOS priors	EOS from GW(+EM)
redback	[15]	Y	dynesty (default), pymultinest, nestle (via bilby, 16 samplers total)	EM transients only (kilonova, GRB, SN)
MOSFiT	[16]	Y	emcee (ensemble MCMC)	EM transients only
jim / flowMC	[17, 18]	Y	Normalizing-flow-enhanced MCMC (JAX/GPU)	GW only
GPU nested sampling	[19]	Y	Nested sampling, GPU-vectorized Slice-within-Gibbs	GW only
dingo	[20, 21]	Y	Amortized neural posterior estimation	GW only
nessai	[22, 23]	Y	Nested sampling with normalizing-flow proposals	GW only

Table 1: Codes performing Bayesian inference on multimessenger-adjacent sources. The top block is directly comparable to NMMA’s problem class (multi-channel or EOS inference); the bottom block is GW-only but included because it is the same likelihood family (compact-binary waveform models) and shows what happens once nested sampling is accelerated or replaced.

Two observations stand out. First, **bajes** [8, 9] is the closest thing to a genuine NMMA competitor: it performs joint, coherent GW+kilonova inference, is open source, and — checked directly in the source of its 2024 joint-inference paper — uses the **dynesty** nested sampler (having used **cpnest** in earlier work). No open-source code was found that performs joint GW+EM+EOS inference together the way NMMA does; NEoS and the Landry/Essick/Legred/Chatziioannou nonparametric-EOS lineage handle the EOS side alone. Second, and more importantly, **every one of these directly-comparable codes is built on nested sampling**: dynesty, MultiNest,

cpnest, or nestle. The multimessenger and nuclear-EOS subfield has not yet adopted any of the alternatives described in Section 3, even though the pure-GW subfield next door already has.

3 Where the neighbouring GW community has already moved, with numbers

The most relevant lesson does not require leaving gravitational-wave astronomy at all: GW parameter estimation has the same likelihood structure as NMMA’s GW branch (an expensive, smooth, differentiable-in-principle waveform likelihood over ~ 10 – 15 parameters), and it has already been substantially accelerated away from plain CPU-bound nested sampling. Every number below was checked directly against the paper’s own $\text{\LaTeX}$ source (fetched via arxiv.org/e-print and grepped), not taken from an abstract.

- **jim / flowMC** [17, 18]: a JAX-based, GPU-native, normalizing-flow-enhanced MCMC sampler. Verified in the jim paper’s source: for GW150914 and GW170817, jim produces ~ 2500 and ~ 3500 effective samples respectively, with a total wall time on a single NVIDIA A100 GPU of **~ 3 minutes**, of which only ~ 40 s is the actual sampling (the rest is one-off JIT-compilation overhead). The same paper reports the equivalent **parallel-bilby** [24]+**dynesty** run, executed on **400 Intel Skylake CPU cores**, taking ~ 2 h for GW150914 and ~ 1 day for GW170817. That is a $\gtrsim 40\times$ to $\gtrsim 400\times$ wall-clock reduction using one GPU instead of hundreds of CPU cores, on the same data and priors.
- **GPU-native nested sampling** [19] (co-authored within the group): rather than replacing nested sampling, this keeps the algorithm and its evidence output exactly as-is and instead re-implements its inner Slice-within-Gibbs kernel to run vectorized on GPU, batching waveform generation across chains. Verified directly in the paper’s abstract and body: for *full-duration, uncompressed* long binary-neutron-star signals across a three-detector network — the exact signal class NMMA’s GW branch targets — it delivers well-calibrated posteriors in a **median of twelve minutes on a single GPU** (five minutes sharded across four GPUs), with no approximation to the likelihood. Adding heterodyning/relative-binning compression [25, 26] brings the median down to **89 seconds**, and a precessing-spin, tidal re-analysis of GW170817 itself completes in **~ 2 minutes**. Because this changes nothing about how NMMA’s scientists would interpret the output — still nested sampling, still a Bayesian evidence, still the full physical waveform — it is arguably the lowest conceptual-risk transplant on this list.
- **Evidence estimation with jim + learned harmonic mean** [27] (also co-authored within the group): benchmarks GPU-based normalizing-flow evidence estimation (jim + **harmonic**, 1 GPU) against **parallel-bilby**+**dynesty** (16 CPU cores) on a 4-dimensional and an 11-dimensional GW inference problem. Verified in the paper’s own source: consistent evidence estimates with speedups of $5.4\times$ (4D) and $14.8\times$ (11D) — the speedup *grows* with dimensionality, the opposite of nested sampling’s usual scaling, and exactly the regime NMMA’s joint high-dimensional parametrizations live in.
- **dingo** [20, 21]: amortized neural posterior estimation — pay a one-off training cost (verified in the PRL paper’s source: “*training takes roughly 10 days*” on a GPU), then obtain a full posterior for a *new* event in seconds. The paper’s own text states inference time is reduced “*from $\mathcal{O}(\text{day})$ to 20 seconds per event,*” cross-validated against **bilby**+**dynesty** and LALInference nested-sampling posteriors via Jensen–Shannon divergence.

- **nessai** [22, 23] takes a hybrid route: keep nested sampling but replace the inefficient rejection-based live-point proposal with a normalizing-flow-informed one, staying entirely inside the framework NMMA already depends on.

The common thread: a smooth, repeatedly-evaluated, GPU-batchable, or learnable likelihood surface is exactly what plain nested sampling never exploits, and what GPU-vectorized nested sampling, flow-enhanced MCMC, amortized neural posterior estimation, and flow-informed proposals all do, in four increasingly aggressive ways.

4 Learning from communities further away

The multimessenger/nuclear-astro community is not alone in having started from MCMC/nested-sampling defaults built for modest dimensionality and cheap-ish likelihoods. Other communities solved essentially the same scaling problem earlier, and each embodies a different strategy that maps onto a different piece of NMMA’s likelihood stack.

Cosmology — surrogate the slow physics, not (only) the sampler.

CMB/large-scale-structure inference (Cobaya [28], CosmoSIS [29]) faces an expensive deterministic solve (a Boltzmann code) inside every likelihood evaluation — the direct analogue of NMMA’s on-the-fly TOV solve for the EOS, or waveform generation for GW. The community’s answer was largely *not* a fancier sampler but a fast, differentiable emulator substituted for the slow solver, after which ordinary gradient-based samplers become affordable. For NMMA, the direct analogue is investing in JAX-differentiable or neural-emulated TOV/EOS and waveform evaluations, rather than only chasing sampler-level speedups.

Probabilistic programming — go gradient-based.

Outside astrophysics, high-dimensional Bayesian inference is dominated by Hamiltonian Monte Carlo / NUTS via **Stan**, **PyMC**, or **NumPyro** [30]. These scale far better with dimension than nested sampling because proposal efficiency does not degrade combinatorially with parameter count, but they require a differentiable log-likelihood. This is the real prerequisite shared by jim and the emulator route above: a JAX (or otherwise autodiff-compatible) rewrite of the likelihood is the single change that unlocks HMC/NUTS, flow-based MCMC, and gradient-informed nested sampling simultaneously.

Simulation-based inference — amortize across events.

The SBI paradigm that dingo imports into GW astronomy (train once on simulated parameter→data pairs, infer new events near-instantly), reviewed broadly in [31] and used heavily for LHC inference via tools such as **MadMiner** [32], is a natural fit for NMMA’s EM branch: kilonova and GRB afterglow simulators are not free to evaluate, but they *can* be called offline ahead of time. The repeated per-event nested-sampling cost NMMA currently pays for every new kilonova/GRB fit is exactly the cost SBI amortizes away.

5 Recommendations

1. **NMMA is not behind the curve within its own subfield.** No comparable open multimessenger or nuclear-EOS code (bajes, redback, MOSFiT, NEoS, the nonparametric-EOS lineage) has moved past nested sampling either. The opportunity is to be first, not to catch up.

2. **Most direct near-term transplant, already in-house:** evaluate the GPU-native nested-sampling kernel of [19] on NMMA’s GW likelihood. Unlike every other item on this list, it changes nothing about the inference paradigm NMMA’s science depends on — still nested sampling, still an evidence estimate, still the full physical waveform — only where the likelihood is evaluated. It was demonstrated on exactly NMMA’s target signal class (long-duration, three-detector BNS) and already has an author in common with the group, making it plausibly the lowest-effort, highest-confidence win on this list.
3. **Lowest-risk incremental step:** add `nessai` [22] as a sampler backend alongside `pymultinest`/`dynesty`/`ultranest`. It is a change to the sampler string, not the likelihood code, and stays inside the nested-sampling/evidence framework NMMA already depends on.
4. **Highest-leverage step, also building on in-house work:** extend the `jim`/`flowMC` + learned-harmonic-mean pipeline, already validated for pure GW parameter estimation and evidence computation by the group itself [27, 17], to NMMA’s joint GW+EM+EOS likelihood. The verified $5.4\times$ – $14.8\times$ (evidence, single-GW, growing with dimension) and $\sim 40\times$ – $400\times$ (full PE, GW150914/GW170817) speedups, on problems of comparable or lower dimensionality than NMMA’s joint parametrization, suggest large, quantifiable gains rather than aspirational ones.
5. **Medium-term, highest structural payoff:** pursue differentiable (JAX-based) or neural-emulated versions of NMMA’s two slow deterministic solves — the GW waveform and the TOV/EOS integration — following the cosmology community’s emulator strategy. This is the change that would make gradient-based samplers (HMC/NUTS) and amortized SBI available across *all* of NMMA’s source types at once, rather than as piecemeal per-messenger fixes.
6. **Longer-term, for the EM branch specifically:** evaluate an amortized/SBI approach (dingo-style neural posterior estimation) for kilonova and GRB fitting, where the same light-curve model is currently re-fit from scratch, at full nested-sampling cost, for every new event.

Methodology note

All literature identification and citation metadata in this document were retrieved programmatically from the INSPIRE-HEP Literature API (<https://inspirehep.net/api/literature>) and are recorded as full BibTeX entries in `references.bib`. Every performance and sampler claim attributed to `jim`, the GPU-native nested-sampling kernel, the Polanska et al. evidence pipeline, `dingo`, and `redback`/`bajes` was verified by fetching the paper’s own L^AT_EX source from arXiv (arxiv.org/e-print/<id>) and grepping the relevant passages directly, rather than relying on abstracts. This process also caught and corrected two errors surfaced during drafting: `redback`’s third supported sampler is `nestle`, not `nessai`, and the parallel-bilby citation key is `Talbot:2019okv`, not `Smith:2019ucc`. NMMA’s own sampler configuration was verified directly from its source code (`nmma/core/parsing.py`, `pyproject.toml`, `nmma/eos/tov.py`) rather than assumed from its publications.

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